

| Location in MLS Chapter 2 of important concepts and equations                                                           |             |                     |
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| Concept                                                                                                                 | page number | Acrobat Reader page |
| $SO(3)$ Special Orthogonal group                                                                                        | 24          | 42                  |
| $so(3)$ Skew symmetric $3 \times 3$ matrices                                                                            | 27, 28      | 45, 46              |
| Lemma 2.3 on $\hat{a}^2$ & $\hat{a}^3$                                                                                  | 28          | 46                  |
| Rodrigues' Formula eqn. (2.14)                                                                                          | 28          | 46                  |
| To calculate $\omega$ and $\theta$ from $R$ : eqn.s (2.17) & (2.18)<br>(essentially the inverse of Rodrigues' Formula)  | 30          | 48                  |
| $SE(3)$ Special Euclidean Group                                                                                         | 35          | 53                  |
| Homogeneous representation of $g \in SE(3)$                                                                             | 36          | 54                  |
| Composition rule for rigid body transformations eq. (2.24)                                                              | 37          | 55                  |
| $se(3)$ (twist and twist coordinates)                                                                                   | 40, top 41  | 58, top 59          |
| Generalized $\vee$ and $\wedge$ operators                                                                               | 41          | 59                  |
| To easily calculate $e^{\hat{\xi}\theta}$ , eqn. (2.36)                                                                 | 42          | 60                  |
| To calculate $e^{\hat{\xi}\theta}$ from $g \in SE(3)$ proposition 2.9                                                   | 42, 43      | 60, 61              |
| Screw coordinates of a twist                                                                                            | 47          | 65                  |
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| Lemma 2.13: $(g\hat{\xi}g^{-1})^\vee = Ad_g\xi$                                                                         | 56          | 74                  |
| $\hat{V}^s$ and $V^s$ eqn. (2.59)                                                                                       | 56          | 74                  |
| $\hat{V}^b$ and $V^b$ eqn. (2.60)                                                                                       | 56          | 74                  |
| $V^s = Ad_g V^b$ eqn. (2.61)                                                                                            | 56          | 74                  |
| $\hat{V}_{ab}^s = \hat{\xi}\dot{\theta}$ and $\hat{V}_{ab}^b = \left(Ad_{g_{ab}^{-1}(0)}\xi\right)^\wedge \dot{\theta}$ | 58          | 76                  |
| Proposition 2.14: $V_{ac}^s = V_{ab}^s + Ad_{g_{ab}} V_{bc}^s$                                                          | 58          | 76                  |
| Proposition 2.15: $V_{ac}^b = Ad_{g_{bc}^{-1}} V_{ab}^b + V_{bc}^b$                                                     | 59          | 77                  |
| Lemma 2.16                                                                                                              | 61          | 79                  |
| Transformation of wrenches eq. (2.67)                                                                                   | 63          | 81                  |
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